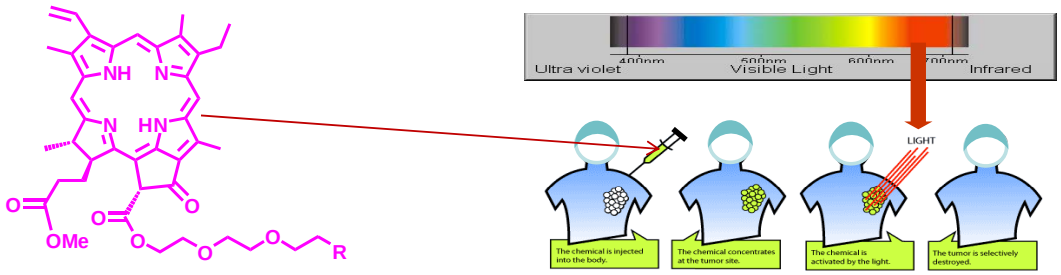
4.3 광역학 치료법

- 광역학 치료법 (Phtodynamic theraphy, PDT): 광감각제 (photosensitizer)를 이용하여 반응성이 매우 좋은 단일항 산소를 생성하고, 이를 이용하여 표적 세포의 기능을 파괴하는 치료법

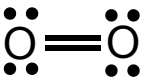
Absorption:	$^1P + h\nu \longrightarrow {}^1P^*$
Intersystem crossing:	$^1P^* \longrightarrow {}^3P^*$
Photosensitization:	$^3P^* + {}^3O_2 \longrightarrow P + {}^1O_2^*$
Oxidation reaction:	$^1O_2^* + \text{reactants} \longrightarrow \text{products}$





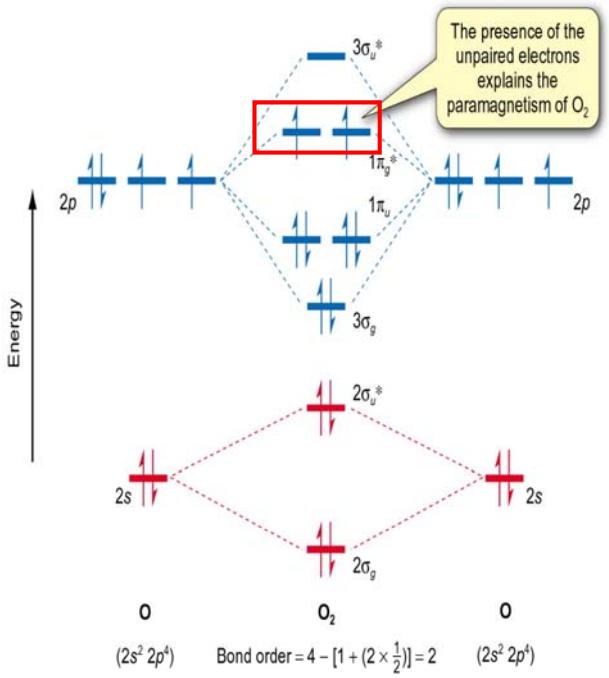
O_2는 상자기성인가 반자기성일까?

Lewis 구조식에 의하면 O_2는 짝짓지 않은 전자를 가지고 있지 않음. → **반자기성 (?)**





O_2 가 상자기성임을 보여주는 실험

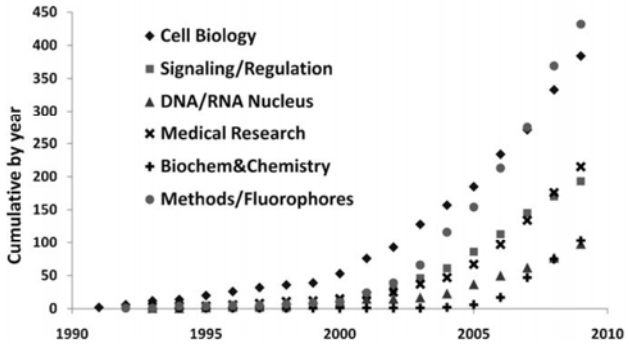


4.4 Fluorescence resonance energy transfer (FRET)

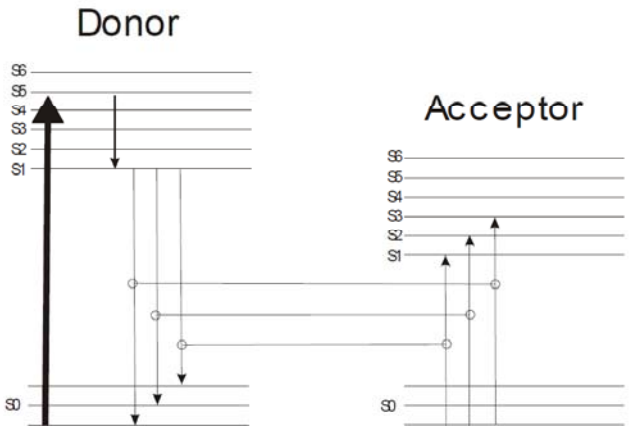
- Fluorescence resonance energy transfer (=Förster resonance energy transfer, FRET): 두 개의 형광 분자 사이에 짧은 거리 (10 nm 미만)에서 일어나는 에너지 이동 현상.

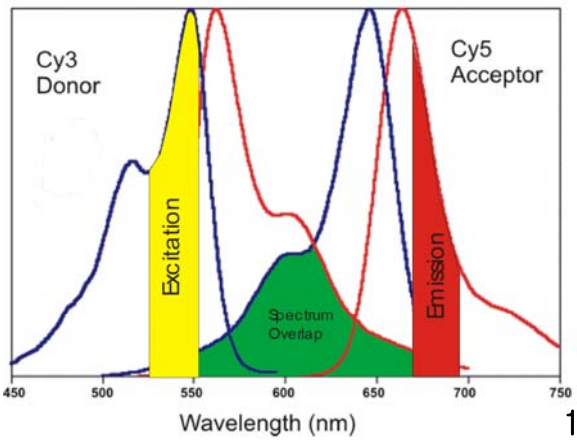


- 1946년에 Theodor Förster에 의해 FRET 현상이 발표 됨.
- Cumulative number of FRET publications by year in broad categories



FRET mechanism



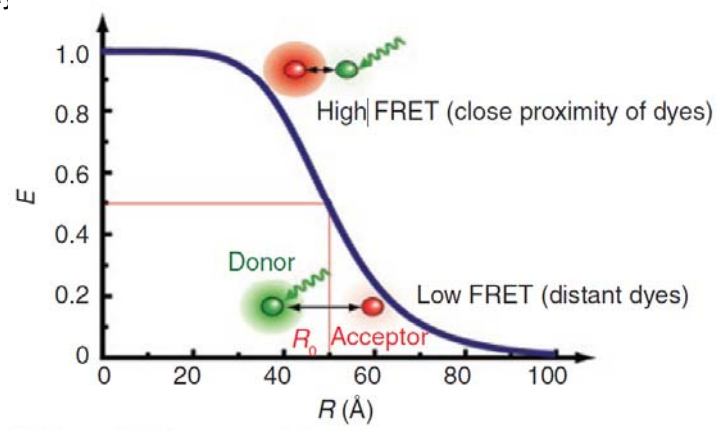


FRET efficiency

- FRET efficiency (E_T)

$$E_T = 1 - \frac{\Phi_f^D}{\Phi_{f,0}^D} = \frac{k_{FRET}}{k_{FRET} + k_R + k_{NR}} = \frac{1}{1 + (k_R + k_{NR})/k_{FRET}} = \frac{1}{1 + (R/R_0)^6}$$

  - k_{FRET}: FRET transfer rate
  - k_R: radiative decay rate
  - k_{NR}: nonradiative decay
  - R_0: Förster distance



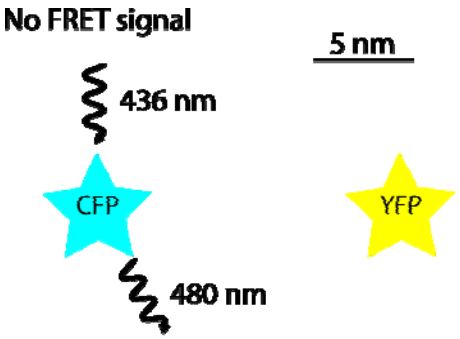
Förster distance

Table 1. Förster distances R_0 and photophysical properties for FP-based FRET pairs.

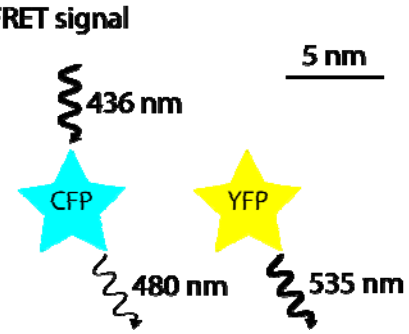
Donor–acceptor[a]	R_0[b] [nm]	FP photophysical properties[c] FP	QY	EC	Ex.	Em.
CFP–YFP	4.755	CFP	0.4	32.5	433	475
Cerulean–Venus	5.215	Cerulean	0.62	43	433	475
mTFP–Venus	5.701	mTFP	0.85	64	462	492
mTFP–mKO2	5.29	GFP	0.6	56	488	507
mTFP–tdTomato	6.201	YFP	0.61	83.4	514	527
mTFP–mTagRFP	5.058	Venus	0.57	92.2	515	528
mTFP–mKate	4.559	mOrange	0.69	71	548	562
GFP–mOrange	5.381	mKO2	0.62	63.8	551	565
GFP–mCherry	5.119	tdTomato	0.69	138	554	581
Venus–tdTomato	6.354	mTagRFP	0.41	81	558	584
Venus–mCherry	5.52	mCherry	0.22	72	587	610
Venus–mKate	5.042	mKate	0.33	45	589	636

Cyan fluorescence protein (CFP) – yellow fluorescence protein (YFP) FRET

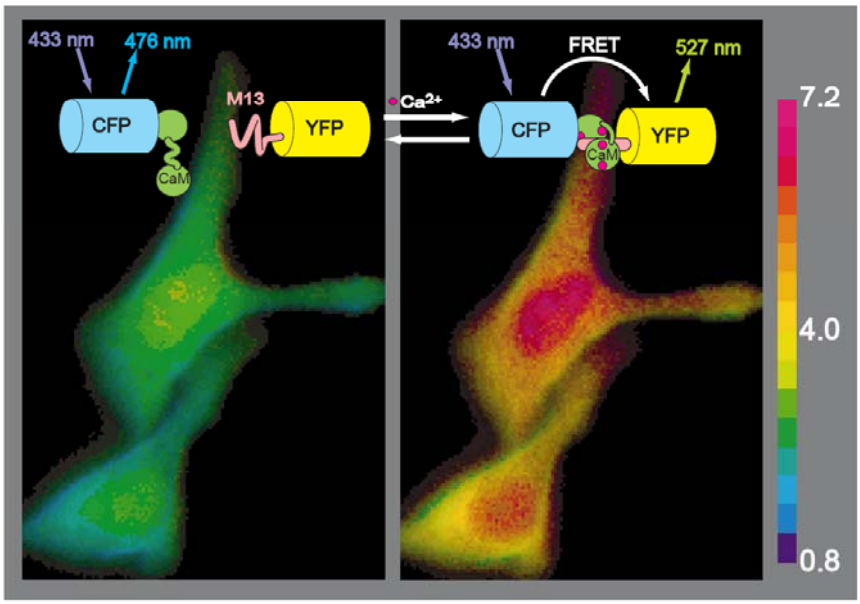
• CFP와 YFP의 거리가 5 nm 이상일 때



• CFP와 YFP의 거리가 5 nm 이하일 때



FRET induced by Ca^{2+} binding to calmodulin



Cyan fluorescence protein (CFP) labeled calmodulin and yellow fluorescence protein (YFP) labeled calmodulin binding peptide (M13–YFP) were coexpressed. High Ca^{2+} levels (right) lead to binding and FRET emission of YFP (pseudo color red); low Ca^{2+} levels (left) lead to little FRET and mostly blue emission (pseudocolor green).

Tsien & Miyawaki, Science **280**, 1954–1955 (1998).